2.Android Resources: Colors, Themes, Strings, Drawables, Dimensions, Images

AIM: To study and implement Android Resources such as Color, Theme, String, Drawable, Dimension, and Image in an Android application.

THEORY:

Android resources are external values used in an app to define UI components without hardcoding. They help maintain consistency, reduce errors, and support localization.

Types of Android Resources Used in This Practical

1. Color Resource (colors.xml)
   Stores all color codes for use across the application.
2. String Resource (strings.xml)
   Stores text messages, titles, labels, etc., making localization easier.
3. Drawable Resource (drawable/)
   Stores images (PNG, JPG, WebP) and XML shapes (rectangles, circles, gradients).
4. Dimension Resource (dimens.xml)
   Used to store sizes such as padding, margin, text size.
5. Theme/Style Resource (styles.xml)
   Defines consistent UI styles.
6. Image Resource
   Images stored under drawable folders.

1. colors.xml

File: res/values/colors.xml

```
<resources>
   <color name="appPrimary">#6200EE</color>
   <color name="appSecondary">#03DAC5</color>
   <color name="white">#FFFFFF</color>
   <color name="black">#000000</color>
</resources>
```

2. strings.xml

File: res/values/strings.xml

```
<resources>
   <string name="app_name">Resource Demo App</string>
   <string name="hello_text">Hello World using Android Resources!</string>
</resources>
```

3. dimens.xml

File: res/values/dimens.xml

```
<resources>
   <dimen name="text_large">24sp</dimen>
   <dimen name="padding_normal">20dp</dimen>
   <dimen name="image_size">150dp</dimen>
</resources>
```

4. styles.xml (Theme)

File: res/values/styles.xml

```
<resources>
   <style name="AppTheme" parent="Theme.MaterialComponents.DayNight">
      <item name="colorPrimary">@color/appPrimary</item>
      <item name="colorAccent">@color/appSecondary</item>
      <item name="android:padding">@dimen/padding_normal</item>
   </style>
</resources>
```

5. Drawable Image

Place any image (e.g., logo.png) inside:
app → res → drawable/

6. Drawable Shape (rounded_box.xml)

File: res/drawable/rounded_box.xml

```
<shape xmlns:android="http://schemas.android.com/apk/res/android"
   android:shape="rectangle">
   <solid android:color="@color/appSecondary" />
   <corners android:radius="12dp" />
</shape>
```

7. Layout File (activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:gravity="center"
   android:padding="@dimen/padding_normal">
   <ImageView
      android:id="@+id/imageView"
      android:layout_width="@dimen/image_size"
      android:layout_height="@dimen/image_size"
      android:src="@drawable/logo" />
   <TextView
      android:id="@+id/textView"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="@string/hello_text"
      android:textColor="@color/appPrimary"
      android:textSize="@dimen/text_large"
      android:paddingTop="20dp" />
   <Button
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Click Me"
      android:background="@drawable/rounded_box"
```

```
        android:textColor="@color/black"
        android:padding="10dp"/>
</LinearLayout>
```

8. Java File (MainActivity.java)

```
package com.example.resourcedemo;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

